

Problem 1:**(b)**

Theorem. Let R be a domain which is a finite type k -algebra for a field k . Then for any ideal $I \subset R$, we have

$$\dim R = \dim R/I + \operatorname{codim} I$$

Proof.

Let $\mathfrak{p} \in \operatorname{Spec}(R)$ give a maximal chain with $\mathfrak{p}$

$$(0) = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r = \mathfrak{p} \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

and by (a) we get that $n = \dim R$ for all of $\mathfrak{p}$'s maximal chains.

Consider $\mathfrak{p}_r \subsetneq \cdots \subsetneq \mathfrak{p}_n$

This can also be considered a chain in $R/\mathfrak{p}$.

Thus, $\dim R/\mathfrak{p} \geq \dim R - \operatorname{codim} \mathfrak{p}$.

Now let $0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_r$ be a max chain in $R/\mathfrak{p}$ s.t. $\dim R/\mathfrak{p} = r$

This corresponds to a chain $\mathfrak{p} \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ in R .

Now extend this chain downwards s.t. we achieve $\mathfrak{p}$'s codimension. Thus, we get a chain of length $r + \operatorname{codim} \mathfrak{p}$.

Thus, $\operatorname{codim} \mathfrak{p} + r \leq \dim R$.

Therefore, we get that $\dim R = \dim R/\mathfrak{p} + \operatorname{codim} \mathfrak{p}$ for $\mathfrak{p}$ prime.

Now let I be an ideal in R . Since R is noetherian we know that I has finitely many minimal prime ideals over it.

Thus, $\dim R/I = \max\{\dim R/\mathfrak{p} \mid \mathfrak{p} \text{ is minimal over } I\}$

Also, by definition of codimension we have $\operatorname{codim} I = \{\operatorname{codim} \mathfrak{p} \mid \mathfrak{p} \text{ is minimal over } I\}$.

Thus, applying the lemma above to each $\mathfrak{p}$ we get that $\dim R/\mathfrak{p} = \dim R - \operatorname{codim} \mathfrak{p}$.

Thus, $\dim R/I = \max\{\dim R/\mathfrak{p} \mid \mathfrak{p} \text{ is minimal over } I\}$

$= \max\{\dim R - \operatorname{codim} \mathfrak{p} \mid \mathfrak{p} \text{ is minimal over } I\} = \dim R - \min\{\operatorname{codim} \mathfrak{p} \mid \mathfrak{p} \text{ is minimal over } I\}$.

Thus, $\dim R/I = \dim R - \operatorname{codim} I$. ■

Problem 2:**(a)**

Compute the dimension of $\mathbb{Q}[x, y, \sqrt{x^7 + y^2}]$.

Proof.

Since $R = \mathbb{Q}[x, y, \sqrt{x^7 + y^2}] \cong \mathbb{Q}[x, y, z]/(z^2 - x^7 - y^2)$. This implies R is a finite type $\mathbb{Q}$ -algebra. Thus, by problem 1 we get that $\dim \mathbb{Q}[x, y, z] = \dim R + \text{codim}(z^2 - x^7 - y^2)$. Since $(z^2 - x^7 - y^2)$ is irreducible this implies it is prime, and this also implies that no ideal but the zero ideal is contained in this. Therefore, since $\mathbb{Q}[x, y, z]$ is a domain this implies that $(z^2 - x^7 - y^2)$ has codimension 1.

Therefore, $\dim R = 3 - 1 = 2$. ■

(b)

Compute the dimension of the ring $R = \mathbb{Z}[x, y, z, w]/(x^3 + yzw + w^5)$.

Proof.

Since $(x^3 + yzw + w^5)$ is irreducible this implies it is prime. Thus, by the PIT we get that $\text{codim}(x^3 + yzw + w^5) \leq 1$, but since we are in a domain its codimension is one. Thus, shown in class we have that $\dim \mathbb{Z}[x, y, z, w] - \text{codim} I \geq \dim R \implies 4 \geq \dim R$.

Also since $(x^3 + yzw + w^5) \subsetneq (x, w) \subsetneq (2, x, w) \subsetneq (2, x, y, w) \subsetneq (2, x, y, z, w)$ is a chain. This implies that $4 \leq \dim R$. Therefore, this implies $\dim R = 4$. ■

(c)

Let $f \in \mathbb{Q}[x, y, z]$ be a nonzero polynomial. Compute the dimension of the ring $\mathbb{Q}[x, y, z]_f$.

Proof.

Since $\mathbb{Q}[x, y, z]_f \cong \mathbb{Q}[x, y, z, t]/(tf - 1)$.

Again by problem 1 since $(tf - 1)$ is irreducible thus prime by PIT its codimension is at most one, but since we are in a domain this implies it has codimension one. Thus, $\dim \mathbb{Q}[x, y, z]_f = \dim \mathbb{Q}[x, y, z, t] - \text{codim}(tf - 1) = 3$. ■

(d)

Let $R = \mathbb{C}[x, y, z]/(xy, xz)$. Compute the dimension of $R_{\bar{x}}$.

Proof.

Since $\mathbb{R}_{\bar{x}} \cong \mathbb{C}[x, y, z, t]/(xy, xz, tx - 1)$.

Also since x is now a unit in $\mathbb{C}[x, y, z, t]/(xy, xz, tx - 1)$ this implies we can cancel the x 's thus, $\mathbb{C}[x, y, z, t]/(xy, xz, tx - 1) = \mathbb{C}[x, y, z, t]/(y, z, tx - 1) \cong \mathbb{C}[x, t]/(tx - 1)$. Then from a similar proof to (c) we get that $\dim \mathbb{C}[x, t]/(tx - 1) = 1$ since $\dim \mathbb{C}[x, t] = 2$ and $\text{codim}(tx - 1) = 1$. ■

(e)

Compute the codimension of the ideal $(x_1, \dots, x_m) \subset k[x_1, \dots, x_n]$, where $m \leq n$ and k is a field.

Proof.

From problem one since $[x_1, \dots, x_n]$ is a finite type k algebra this implies $\text{codim}(x_1, \dots, x_m) = m$ since $\dim [x_1, \dots, x_n] = n$ and $\dim [x_1, \dots, x_n]/(x_1, \dots, x_m) = n - m$. ■

Problem 3:

This problem is about UFDs from the point of Noetherian rings and dimension theory.

(b)

Prove that a domain R is a UFD if and only if the following conditions holds:

- i. For every nonzero, noninvertible element $x \in R$ there exists a factorization $x = \pi_1 \dots \pi_r$ where $\pi_1, \dots, \pi_r \in R$ are irreducible.
- ii. Every irreducible element in R is prime.

Proof.

Assume R is a UFD.

i. is immediate from the definition of UFD.

Let $x \in R$ be irreducible.

Let $x \mid ab$. Then $ab = xr$ s.t. $r \in R$.

Then factor a, b, c into irreducibles. By the uniqueness of factorizations this implies that x must appear among the irreducible factors of either a or b . Thus, $x \mid a$ or $x \mid b$.

Therefore, x is prime.

Assume *i.* and *ii.*

First existence of factorization exists from *i.*

Suppose $x \in R$ has two factorizations $x = p_1 \dots p_n = q_1 \dots q_m$.

Since $p_1 \mid q_1 \dots q_m$ by *ii.* this implies $p_1 \mid q_j$. Thus, $q_j = up_1$ for some unit u .

Therefore, you can do this over and over which implies $n = m$ and $p_i = u_j q_j$ for units u_j .

Therefore, factorizations are unique up to units.

Thus, R is a UFD. ■

(c)

Prove that if R is a noetherian domain, then for every nonzero, noninvertible element $x \in R$, there exists a factorization $x = \pi_1 \dots \pi_r$ where $\pi_1, \dots, \pi_r \in R$ are irreducible elements. In particular, combined with (b), we see that a noetherian domain is a UFD if and only if every irreducible element is prime.

Proof.

Assume R is a noetherian domain.

Let $S = \{x \in R \setminus \{0\} \mid x \text{ is not a unit, and } x \text{ does not factor into irreducibles}\}$.

Assume for contradiction that $S \neq \emptyset$.

Then consider $\{(x) \mid x \in S\} \neq \emptyset$. Since R is noetherian this implies that there exists a maximal element (y) in this set.

Since $y \in S$ this implies y is not irreducible thus $y = ab$ s.t. $a, b \in R$ and are not units.

Since $(x) \subsetneq (a), (b)$. This implies a, b do factor into irreducibles since (x) is maximal thus, $a, b \notin S$.

Thus, $a = p_1 \dots p_n$ and $b = q_1 \dots q_m$, but this implies $x = p_1 \dots p_n q_1 \dots q_m$ which is a contradiction. Therefore, $S = \emptyset$.

Therefore, all noninvertible elements factor into irreducibles. ■

(d)

Prove that if R is a noetherian domain, then R is a UFD if and only if every prime ideal $\mathfrak{p} \in \text{Spec}(R)$ with $\text{codim} \mathfrak{p} = 1$ is a principal ideal.

Proof.

Assume R is a noetherian domain and a UFD.

Let $\mathfrak{p} \in \text{Spec}(R)$ s.t. $\text{codim} \mathfrak{p} = 1$. Let $\mathfrak{p} = (a, c_1, \dots, c_n)$

Then consider $(a) \subsetneq \mathfrak{p}$. Since R is a domain this implies that (a) is not prime since if it was then we would have $(0) \subsetneq (a) \subsetneq \mathfrak{p}$. Which would implies $\mathfrak{p}$ has codimension 2.

Thus, (a) is not prime. Thus, a is not irreducible this implies $a = p_1 \dots p_n$ s.t. p_i are irreducible. Consider (p_1) which is prime since we are in a UFD. This implies that $(0) \subsetneq (p_1)$ is a chain. Thus, (p_1) has codimension 1 by the PIT, but since $(p_1) \subset \mathfrak{p}$ this implies that $(p_1) = \mathfrak{p}$. Therefore, $\mathfrak{p}$ is a principal ideal. ■

Problem 4:**(a)***Proof.*

Let $f \in k\{\{t\}\}$ s.t. $f = \sum_{i=i_0}^{\infty} a_i t^{i/n}$ for some $i_0 \in \mathbb{Z}$ and $n \in \mathbb{N}$, and $a_i \in k$ and $a_{i_0} \neq 0$.

Suppose f also is equal to $\sum_{j=j_0}^{\infty} b_j t^{j/m}$ for some $j_0 \in \mathbb{Z}$ and $m \in \mathbb{N}$, and $b_j \in k$ and $b_{j_0} \neq 0$.

Let $N = \text{lcm}(n, m)$. Then this implies both can be written as s.t. $i/n = (iN/n)/N$ and same for j/m .

$$f = \sum_{r=r_0}^{\infty} c_r t^{r/N} = \sum_{r=r_0}^{\infty} d_r t^{r/N}$$

This implies $c_r = d_r$ for all r .

Therefore, this implies that the first non zero term for c_r has power on t equal to $i/n = (iN/n)/N$ since these are the same series this implies that $(iN/n)/N = (jN/m)/M \implies i/n = j/m$.

Thus, the valuation map is well-defined.

Let $f \in k\{\{t\}\}$ s.t. $v(f) = \infty$ this implies that there is no nonzero terms in f . Thus, $f = 0$.

If $f = 0$ then this implies that there is no minimal nonzero term thus $v(f) = \infty$.

Let $f, g \in k\{\{t\}\}$ s.t.

$$f = \sum_{i=i_0}^{\infty} a_i t^{i/n}, \quad a_{i_0} \neq 0 \tag{1}$$

$$g = \sum_{j=j_0}^{\infty} b_j t^{j/m}, \quad b_{j_0} \neq 0 \tag{2}$$

Then $fg = \sum_{k=mi_0+nj_0} c_k t^{k/(nm)}$ s.t. $c_k = \sum_{mi+nj=k} a_i b_j$.

Thus, looking at the smallest possible exponent we get $c_{mi_0+nj_0} = a_{i_0} b_{j_0} \neq 0$ since $a_{i_0} \neq 0 \neq b_{j_0}$.

Therefore, $v(fg) = (mi_0 + nj_0)/(nm) = i_0/n + j_0/m = v(f) + v(g)$.

If one of f, g are zero then $v(fg) = \infty = v(f) + v(g)$.

Let $f, g \in k\{\{t\}\}$.

If $g = 0$ then $v(f + g) = v(f) \geq \min\{v(f), \infty\} = v(f)$.

So assume $f, g \neq 0$.

First choose some common denominator for the exponents in the series call this n .

Thus, $f = \sum_{i=i_0} a_i t^{i/n}$ and $g = \sum_{j=j_0} b_j t^{j/n}$.

Let $f + g = \sum_{k=l_0} c_k t^{k/n}$

If $f + g = 0$ then $v(f + g) = \infty \geq \min\{v(f), v(g)\}$

Assume $f + g \neq 0$ then if $v(g) < v(f)$ then $v(f + g) = j_0/n \geq v(g)$.

If $v(g) = v(f)$ Then

if $c_{i_0} \neq 0$ then $v(f + g) = i_0/n \geq i_0/n$

if $c_{i_0} = 0$ then this implies that the first nonzero term has a greater exponent than i_0 .

Thus, $v(f + g) = l_0 > i_0 = j_0$.

Therefore, $v(f + g) \geq \min\{v(f), v(g)\}$

Therefore, v is a valuation map. ■

(b)

Let k be a field and let $k(x, y)$ be the fraction field of $k[x, y]$. Show that there is a valuation $v : k(x, y) \rightarrow \mathbb{Z}^{\oplus 2} \cup \{\infty\}$, where $\mathbb{Z}^{\oplus 2}$ is given the lexicographical order, which is uniquely determined by $v(x^a y^b) = (a, b)$ for $(a, b) \in \mathbb{Z}^{\oplus 2}$.

Proof.

Define $w : k(x, y) \rightarrow \mathbb{Z}^{\oplus 2} \cup \infty$ s.t. $w(0) = \infty$ and for $f \in k[x, y]$ s.t. $f = \sum c_{i,j} x^i y^j$ that $w(f) = (a, b)$ s.t. (a, b) is the min element in $\{(i, j) \in \mathbb{N}^2 \mid c_{i,j} \neq 0\}$.

Now for $f \in k(x, y)$ s.t. $f = g/h$ s.t. $g, h \in k[x, y]$ and $h \neq 0$ that $w(f) = w(g) - w(h)$.

Multiplicativity on $k[x, y]$:

Let $f, g \in k[x, y]$ then consider $w(fg)$ this implies the new minimal nonzero term are just the minimal terms of f, g multiplied together. Thus, if $w(f) = (a, b)$ and $w(g) = (i, j)$ then $w(fg) = (a + i, b + j) = w(f) + w(g)$.

Well-Defined on $k(x, y)$:

Suppose $f/g = h/e$. This implies $fe = hg \neq 0$. Thus, by above we have that $w(fe) = w(hg) \implies w(f) - w(g) = w(h) - w(e)$. Thus, this is well-defined.

Multiplicativity on $k(x, y)$:

$w(f/g \cdot h/e) = w(fh) - w(ge) = (w(f) - w(g)) + (w(h) - w(e)) = w(f/g) + w(h/e)$.

Inequality:

$w(f/g + h/e) = w((fe + hg)/ge) = w(fe + hg) - w(ge)$.

Consider $w(fe + hg)$. Assume fe or hg are not zero since if fe is then we get $w(fe + hg) = w(fe) \geq w(fe)$. Same with if hg is then we get $w(fe + hg) = w(hg) \geq w(hg)$. Same with if $fe + hg = 0$ then $w(fe + hg) = \infty \geq \min\{w(fe), w(hg)\}$.

If $w(fe) < w(hg)$ then $w(fe + hg) = w(fe) \geq \min\{w(fe), w(hg)\}$.

If $w(fe) = w(hg)$ and the minimal term with exponent $w(fe)$ is not zero. Then $w(fe + hg) = w(fe) \geq w(fe)$.

If the term with exponent $w(fe)$ is zero. This implies the minimal nonzero term is

bigger. Thus, $w(fe + hg) \geq \min w(fe)$.

Thus, $w(f/g + h/e) \geq \min\{w(fe) + w(hg)\} - w(ge) = \min\{w(f) + w(e), w(h) + w(g)\} - w(g) - w(e) \geq \min\{w(f) - w(g), w(g) - w(e)\} = \min\{w(f/g), w(g/e)\}$.

Also, if $w(a) = \infty \iff a = 0$ by definition of w .

Now for $x^a y^b$ for $(a, b) \in \mathbb{N}^2$ we get that $v(x^a y^b) = (a, b) = w(x^a, y^b)$.

So for general $a, b \in \mathbb{Z}^{\oplus 2}$ pick A, B large enough s.t. $a + A \geq 0$ and $b + B \geq 0$ thus $x^a y^b = (x^{a+A} y^{b+B}) / (x^A y^B)$.

Thus, $w(x^a y^b) = w(x^{a+A} y^{b+B}) - w(x^A y^B) = (a, b) = v(x^a y^b)$.

uniqueness:

Let v be a valuation on $k(x, y)$ to $\mathbb{Z}^{\oplus 2} \cup \{\infty\}$ with the property $v(x^a y^b) = (a, b)$.

Let $f \in k[x, y]$ s.t. $f \neq 0$.

$$f = \sum c_{i,j} x^i y^j$$

let (a_0, b_0) be the exponents on the minimal degree term with lexicographical order.

This implies that $f = x^{a_0} y^{b_0} c_{(a_0, b_0)} (1 + c_{(a_0, b_0)}^{-1} h(x, y))$ s.t. $h(x, y) \in k[x, y]$.

Consider $v(1 + c_{(a_0, b_0)}^{-1} h(x, y)) \geq v(1) = 0$, also $v(h) > (0, 0)$ since is either zero on a polynomial with a zero constant term since we factored out all the lower degree terms.

Thus, $v(h) > (0, 0)$.

Consider $1 = (1 + h) - h$.

$(0, 0) = v(1) = v(1 + h - h) \geq \min\{v(1 + h), v(h)\}$ but since $v(h) > (0, 0)$ we know that $(0, 0) \geq v(1 + h)$.

Also since $v(1 + h) \leq \min\{v(h), v(1)\} = (0, 0)$. This implies $v(1 + h) = v(0, 0)$.

Thus, $v(1 + c_{(a_0, b_0)}^{-1} h) = (0, 0)$ also since $v(c_{(a_0, b_0)}^{-1} h) = (0, 0) + v(h)$.

So, do the same proof with that therefore, applying v to f we get that.

$$v(f = v(x^{a_0} y^{b_0} c_{(a_0, b_0)} (1 + c_{(a_0, b_0)}^{-1} h(x, y)))) = v(x^{a_0} y^{b_0}) + v(c_{(a_0, b_0)}) + v(1 + c_{(a_0, b_0)}^{-1} h) = (a, b) + (0, 0) + (0, 0) = (a + b).$$

Therefore, $v(f) = w(f)$.

Thus, for $f \in k(x, y)$ s.t. $f \neq 0$.

Then consider $v(1) = v(ff^{-1}) \implies 0 = v(f) + v(f^{-1}) \implies v(f) = -v(f^{-1})$.

Thus, consider $f/g \in k(x, y)$.

This implies $v(f/g) = v(fg^{-1}) = v(f) - v(g) = w(f) - w(g) = w(f/g)$. Therefore, $v = w$. ■

Problem 5:

Let $v' : K \rightarrow \Gamma' \cup \{\infty\}$ be any valuation on K such that $R = \{x \in K \mid v'(x) \geq 0\}$ is the valuation ring of v' . Prove that there is an injective order preserving group homomorphism $\phi : \Gamma \rightarrow \Gamma'$ such that $v'(x) = \phi(v(x))$ for all $x \in K^\times$.

Proof.

Define $\phi : \Gamma \rightarrow \Gamma'$ s.t. $\phi([x]) = v'(x)$.

Well-Defined:

If $[x] = [y]$ in Γ then $x/y, y/x \in R$ thus, $v(x/y), v(y/x) \geq 0$.

Thus, $v'(x) = v'(yxy^{-1}) = v'(y) + v'(xy^{-1}) \geq v'(y)$ and $v'(y) = v'(xyx^{-1}) = v'(x) + v'(yx^{-1}) \geq v'(x)$.

Therefore, $\phi([x]) = v'(x) = v'(y) = v'([y])$.

homomorphism:

Consider $\phi([x] + [y]) = \phi([xy]) = v'(xy) = v'(x) + v'(y) = \phi([x]) + \phi([y])$.

Order:

Let $[x] \leq [y]$ this implies $y/x \in R \implies v'(y/x) \geq 0$.

Thus, $\phi([y]) = v'(y) = v'(xyx^{-1}) = v'(x) + v'(yx^{-1}) \geq v'(x) = \phi([x])$.

Injective:

Let $[x] \in \Gamma$ s.t. $\phi([x]) = 0$.

This implies $v'(x) = 0$ thus $0 = -v'(x) = v'(x^{-1})$. This implies that $x, x^{-1} \in R$. Thus, $x/1 \in R$ and $1/x \in R$. This implies that $[1] = [x]$. Thus, $\phi([x]) = \phi([1]) = v'(1) = 0$.

Thus, injective.

Also since $v(x) = [x]$ this implies $\phi(v(x)) = \phi([x]) = v'(x)$. ■